

A categorical interpretation of possibilistic operators in formal concept analysis

Torgeir Aambø

Norwegian Defence Research Establishment (FFI)

Abstract

In this short paper we describe Dubois–Prades eight possibilistic operators in formal concept analysis as arising from Kan extensions, Kan lifts and De Morgan duals of the underlying relational profunctor. This allows us to complete their description of fixedpoints of asymmetric operator compositions, showing in particular that no other compositions give rise to formal concepts outside of the standard Galois connection and the NII-pairs.

1 Introduction

Formal concept analysis (FCA) is a technique for structuring data by associated attributes [cite](#), and has during recent years seen a wide array of applications in various contexts [cite](#). Given a list of objects, a list of attributes, and a relation describing which object has which attribute, FCA constructs a set of *formal concepts* which are naturally structured into a lattice.

Formal concepts are constructed as fixedpoints of derivation operations $(-)'$ on the set of objects and attributes. More formally, given a set of objects G , a set of attributes M and a relation $R \subseteq G \times M$, one defines for a subset of objects $A \subseteq G$ its *intent*

$$A' = \{m \in M \mid \forall a \in A, aRm\},$$

and similarly for a subset of attributes $B \subseteq M$ its *extent*

$$B' = \{g \in G \mid \forall b \in B, gRb\},$$

where xRy means that the object x has the attribute y . In words, the extent of A is the set of common attributes shared by all of its objects, and the intent of B is the set of objects that have all of its attributes. A formal concept is then a subset $(A, B) \subseteq G \times M$ such that $B = A'$ and $A = B'$, see [cite](#) for details.

In [cite](#), the authors take inspiration from possibility theory, see [cite](#), and introduce additional operators on G and M based on the possibilistic powerset operations introduced by Dubois–Prade in [cite](#). Denoting the previous FCA operator by $(-)' = (-)^\Delta$, they further introduce for a subset $A \subseteq G$ the operators

$$A^\Pi = \{m \in M \mid \exists a \in A, aRm\},$$

as well as

$$A^\nabla = \{m \in M \mid \exists a \notin A, a\bar{R}m\},$$

where $\bar{R} = \neg R$ is the negated relation, and finally

$$A^N = \{m \in M \mid \forall a \in M, aRm \implies a \in A\}.$$

These are usually called the potential possibility Π , the guaranteed necessity N , the guaranteed possibility Δ and the potential necessity ∇ . One can similarly define these four operators for subsets $B \subseteq M$, which we denote with subscripts, $(-)_{\Pi}, (-)_N, (-)_{\Delta}$ and $(-)_{\nabla}$.

By design, fixedpoints of $(-)^\Delta \circ (-)_{\Delta} : P(G) \longrightarrow P(G)$ are exactly the extents of the formal concepts of the formal context (G, M, R) . In [cite](#), the authors prove that the fixedpoints for the asymmetric operator composition $(-)^N \circ (-)_{\Pi}$, which they call NII-pairs, are precisely the formal concepts of the complementary formal context $(G, M, \bar{R})$ defined using the negated relation $\bar{R} = \neg R$. They note at the end of the paper that further research should be done to understand the remaining asymmetric operator compositions.

By using a well established categorical framework for FCA, the first goal of this paper is to prove that the above four operators arise canonically from duals on the underlying relation R , rather than from possibility-theoretic considerations. The second goal is to understand the structure of these operators, prove that their relationships come from simple categorical constructions, and classify the fixedpoints of the remaining asymmetric operator compositions.

2 Category theory and FCA

Formal concept analysis has attracted interest by category theorists for being a natural and useful application of their abstract tools. Category theory was introduced by Eilenberg and MacLane in [cite](#), and is an abstract mathematical theory for understanding compositional structure – see [cite](#) for a comprehensive introduction.

A category consists of *objects* and *morphisms*, which are abstract relationships between objects. Category theory can then be thought of as studying objects by understanding how they relate to all other objects of the same defined type. We will assume that the reader is familiar with basic categorical constructions.

2.1 The closure operators

For a category theorist, the objects G and attributes M can be viewed as discrete categories. A relation between G and M is formalized as a $\mathbf{2} = \{0, 1\}$ -valued profunctor $R: G \nrightarrow M$, see [cite](#) for details. To construct the associated formal concepts, notice that a the profunctor is by currying equivalent to two functors

$$R^-: G^{\text{op}} \longrightarrow [M, \mathbf{2}] \text{ and } R_-: M \longrightarrow [G^{\text{op}}, \mathbf{2}].$$

As $[M, \mathbf{2}]$ is cocomplete and $[G^{\text{op}}, \mathbf{2}]$ is complete, there are unique extensions by (co)Yoneda to functors

$$[G^{\text{op}}, \mathbf{2}] \xrightleftharpoons[R_*]{R^*} [M, \mathbf{2}]^{\text{op}}$$

which can be shown to be adjoints, see [cite](#). As G and M are discrete, the functor categories are precisely the powersets of G and M respectively: $[G^{\text{op}}, \mathbf{2}] = P(G)$ and $[M, \mathbf{2}]^{\text{op}} = P(M)^{\text{op}}$.

These functors are explicitly given by using categorical ends, meaning that

$$R^*(A)(m) = \int_{g \in G} \mathbf{2}(A(g), gRm),$$

and

$$R_*(B)(g) = \int_{m \in M} \mathbf{2}B(m), (gRm)$$

where A and B are presheafs. In $\mathbf{2}$, the internal hom is given by implication, and the end becomes a conjunction

$$R^*(A)(m) = \bigwedge_{g \in G} (A(g) \implies gRm).$$

In other words, for every object g , if g belongs to A , then g has the property m , which is precisely the FCA closure operation $(-)^{\Delta} = (-)'$ on $P(G)$ as earlier. Similarly one gets the FCA closure operation $(-)_{\Delta}$ on $P(M)$.

We can now recognize the formal concepts of (G, M, R) as the fixedpoints of the adjunction $R^* \dashv R_*$, often called the *nucleus* of the profunctor R , see [cite](#). More formally, the composition $R_* \circ R^*$ is a monad on $P(G)$, and the extents of the formal concepts are the objects in the Eilenberg–Moore category of this monad. Similarly, taking opposite categories gives a monad $R^{*, \text{op}} \circ R_*^{\text{op}}$ on $P(M)$, and its Eilenberg–Moore category gives the intents of the formal concepts. The nucleus consists of pairs of such objects and attributes, i.e., an object in $\text{Nuc}(R)$ is of the form (A, B) where $A = R_*(R^*(B))$ and $B = R^{*, \text{op}}(R_*^{\text{op}}(A))$. By adjunction, however, these determine each other, so we can focus only on one of them.

To summarize, the formal concepts arise completely naturally via simple categorical constructions from the profunctor R , via the fixedpoints of the monad $R_* R^*$ on $[G^{\text{op}}, \mathbf{2}]$.

2.2 Idempotency and closure

The operators defining FCA are so-called closure operators on posets. By definition a closure operator is extensive, monotone and idempotent. In the case of an adjunction from a formal context as described in the previous sections, this means that

1. $A \subseteq R_* R^*(A)$,
2. $A \subseteq A' \implies R_* R^*(A) \subseteq R_* R^*(A')$, and
3. $R_* R^* R_* R^*(A) = R_* R^*(A)$, respectively.

Any adjunction coming from a profunctor from a formal context satisfies all three of these. By [cite](#), the monad $R_* R^*$ is idempotent, and the two other properties follow formally from being $\mathbf{2}$ -functors. In fact, any monadic functor on a poset satisfies these, see [cite](#).

In order to be recognized as a FCA closure operator on *some* formal context, these three axioms have to be fulfilled, otherwise the resulting fixedpoints are not formal concepts. The following is rather trivial then, but we state it for reference later.

Proposition 2.1. *Let $L: \mathcal{C} \rightarrow \mathcal{C}$ be an endofunctor on a poset. If L fails to be extensive, monotone or idempotent, then there is no formal context (G, M, R) such that $\text{EM}(L) \simeq \text{Nuc}(R)$.*

This result will help us quickly identify which of the asymmetric compositions later possibly could give rise to alternative formal concepts on some related formal context.

Remark 2.2. Dually, given an adjoint pair $L \dashv R$, the composition LR is a comonad. On posets, this comonad satisfies dual properties of the closure operators, called being an *interior* operator. In the case of formal concepts, instead of completing a subset A to a formal concept, i.e., constructing the smallest formal concept containing the set, the interior operator finds the largest formal concept *inside* A .

3 The other operators

One way to define the functor R^* on $P(G)$ as we did earlier is to recognize it as a right Kan extension. This prompts the idea to also consider the *left* Kan extension, or equivalently the coend-construction instead of the end-construction, as well as the right and left Kan lifts. These, together with the negation operation on the relation R , will give us all eight operations described by Dubois–Prade.